

Zifeng Zhang

New Haven, CT
zifeng.zhang@yale.edu
nk12chang@gmail.com

jornzhang.github.io

Academic Appointments

Postdoctoral Associate	2025–Present
Department of Biostatistics, Yale School of Public Health, New Haven, CT	
Postdoctoral Mentor: Prof. Hongyu Zhao	

Education

Ph.D. in Statistics , Colorado State University	2025
Dissertation: <i>New developments on linear regression with random design and high-dimensional mediation analysis</i>	
Advisor: Prof. Wen Zhou; Co-advisor: Prof. Haonan Wang	
M.A.S. in Applied Statistics , Nankai University, Tianjin, China	2018
B.A. in Economics, Specialization in Insurance , Nankai University, Tianjin, China	2016

Research Interests

- High-dimensional statistics
- Robust statistical inference under complex dependence
- Causal inference and high-dimensional mediation analysis
- Statistical learning with embeddings and biomedical applications

Peer-Reviewed Publications

1. **Zhang, Z.**, Ding, P., Zhou, W., and Wang, H. (2025). “With Random Regressors, Least Squares Inference Is Robust to Correlated Errors with Unknown Correlation Structure.” *Biometrika*, 112(1), asae054. [[Journal](#)] [[arXiv](#)]

Working Papers and Manuscripts in Preparation

2. **Zhang, Z.**, Lei, L., Ding, P., and Zhou, W. “Two Routes to Validity: Why Nonstandard Standard Errors Are More Robust Than You Think.” Manuscript in preparation.
3. **Zhang, Z.**, Yang, F., Ding, P., and Zhou, W. “High-Dimensional Mediation with Treatment–Mediator Interactions: Signal-Induced Ill-Conditioning and Inference.” Manuscript in preparation.
4. **Zhang, Z.**, Ding, P., and Zhou, W. “A Duality of the F Distribution.” Manuscript in preparation.
5. **Zhang, Z.**, Wang, L., Huo, Z., Wu, P., Liu, Y., Liu, D., and Zhao, H. “Enhancing Prognostic Accuracy in Idiopathic Pulmonary Fibrosis: A Novel Momentum Feature for Longitudinal Biomarker Analysis.” Manuscript in preparation.

Research in Progress

6. **Zhang, Z.**, Tian, Y., Feng, Y., and Zhao, H. “Embedding-Assisted Elastic Net.”

7. Magid-Bernstein, J., **Zhang, Z.**, Grychowski, L., Hadiwidjaja, B., Sorensen, G., Remillard, W., Lidman, A., Vargas, D., Falcone, G. J., Sheth, K. N., Gilmore, E. J., Matouk, C., Petersen, N., Lyman, K., O’Keefe, L., Kim, J., Zhao, H., and Sansing, L. H. “Distinct Cytokine Trends Are Associated with Outcome after Aneurysmal Subarachnoid Hemorrhage.”

Teaching Experience

Instructor of Record, Colorado State University

STAT 301: Introduction to Applied Statistics Fall 2021–Spring 2025 (8 semesters)

Undergraduate introduction to descriptive statistics, statistical inference, ANOVA, and simple and multiple linear regression.

STAT 201: General Statistics Summer 2020, Summer 2024

Undergraduate introduction to descriptive statistics, statistical inference, and simple linear regression.

Recitation Instructor, Colorado State University

STAT 201: General Statistics Fall 2019, Spring 2020, Spring 2021

Teaching Assistant, Colorado State University

STAT 730: Advanced Theory of Statistics Spring 2023, Spring 2024, Spring 2025

STAT 720: Probability Theory Fall 2022, Fall 2023, Fall 2024

STAT 520: Introduction to Probability Theory Fall 2021

STAA 573: Analysis of Time Series Fall 2020

STAA 551: Regression Models and Applications Fall 2020

STAT 315: Intro to Theory and Practice of Statistics Fall 2019

Invited Talks

- “Dual Validity in Linear Regression: Rethinking Robust Inference When Error Structures Are Unknown.” ASA Section on Statistical Learning and Data Science Conference (SLDS), New York University, New York, NY, November 2026. *Invited talk, forthcoming.*
- “Dual Validity in Linear Regression: Rethinking Robust Inference When Error Structures Are Unknown.” New England Statistics Symposium (NESS), University of Connecticut, Storrs, CT, May 2026. *Invited talk.*
- “Dual Validity in Linear Regression: Rethinking Robust Inference When Error Structures Are Unknown.” New Jersey Institute of Technology, Newark, NJ, December 2025. *Statistics seminar.*

Contributed Presentations

- “When Least Squares Inference with Random Regressors Encounters Unknown Correlated Errors.” Western North American Region Conference (WNAR), Fort Collins, CO, June 2024. *Student paper oral presentation.*
- “Mediation Analysis with Interactions and Heredity under Growing Number of Mediators.” Institute for Research in Statistics and its Applications (IRSA), Minneapolis, MN, May 2024. *Student Poster Competition.*

Academic Service

Journal Referee

Biometrika

Organizer

Zhao Lab Journal Club, Yale School of Public Health, Summer 2026

Honors and Awards

Graduate Scholarship for Outstanding Freshmen

Nankai University, China 2016

Gongneng Scholarship

Nankai University, China 2013

Professional Memberships

New England Statistical Society (NESS), Regular Member

2026–Present

Computing Skills

R; $\text{\LaTeX}$; Slurm; high-performance and parallel computing